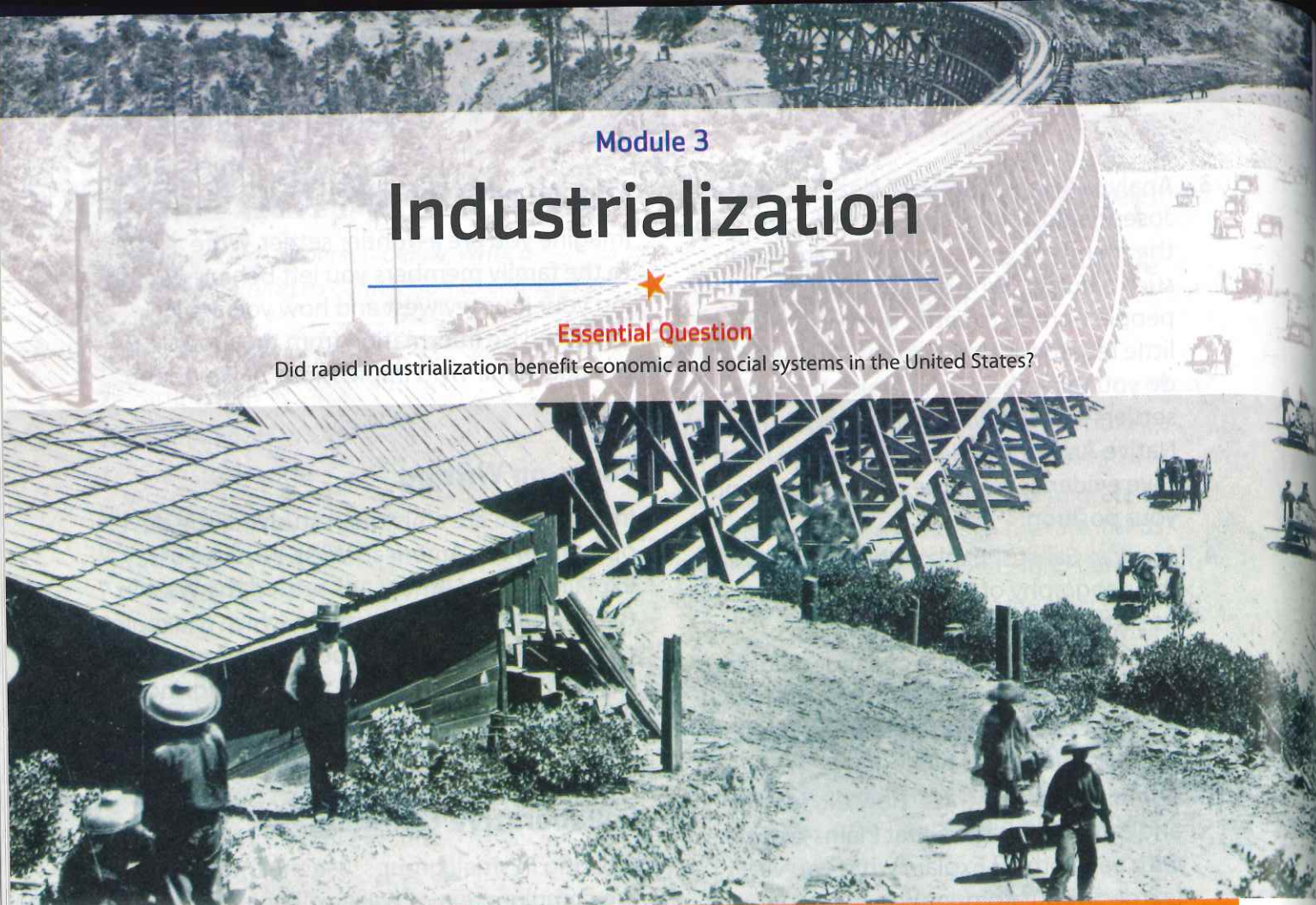


Industrialization

Essential Question

Did rapid industrialization benefit economic and social systems in the United States?



About the Photograph: Chinese laborers from the Central Pacific Railroad blast tunnels and construct bridges in a race to lay track through the Sierra Nevada. Blasting tunnels was difficult; sometimes laborers progressed less than a foot a day.

In this module you will learn about the industrial revolution of the late 1800s, sometimes referred to as the second industrial revolution, and the impact it had on American businesses and workers.

What You Will Learn . . .

Lesson 1: The Expansion of Industry 154

The Big Idea At the end of the 19th century, natural resources, creative ideas, and growing markets fueled an industrial boom.

Lesson 2: The Age of Railroads 162

The Big Idea The growth and consolidation of railroads benefited the nation but also led to corruption and required government regulation.

Lesson 3: Big Business 169

The Big Idea The expansion of industry resulted in the growth of big business.

Lesson 4: The Rise of the Labor Movement 176

The Big Idea The expansion of industry prompted laborers to form unions to better their lives.

Explore ONLINE!



VIDEOS, including...

- Great Minds in Business: Andrew Carnegie
- Oil
- Setting Time Zones
- The Rise of J. P. Morgan
- Traits of a Titan
- Homestead Strike

- ✓ Document-Based Investigations
- ✓ Graphic Organizers
- ✓ Interactive Games
- ✓ Animation: Early Refrigerated Railroad Cars
- ✓ Image with Hotspots: Triangle Shirtwaist Fire

Timeline of Events 1868–1901

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United States Events

World Events

1868

1869 Central Pacific and Union Pacific complete the transcontinental railroad.

1870 Franco-Prussian War breaks out.

1875 British labor unions win right to strike.

1876 Alexander Graham Bell invents the telephone.

1877 *Munn v. Illinois* establishes government regulation of railroads. Mother Jones supports the Great Strike of 1877.



< 1879 Thomas A. Edison invents a workable light bulb.

1882 United States restricts Chinese immigration. ^

1883 Germany becomes the first nation to provide national health insurance.

1884 Grover Cleveland is elected president.



< 1886 Haymarket Riot turns public sentiment against unions.

1890 Congress passes the Sherman Antitrust Act.

1890 Colonization of sub-Saharan Africa peaks.

1893 Women in New Zealand gain voting rights.

1894 President Cleveland sends federal troops to Illinois to end the Pullman strike.

1896 William McKinley is elected president.

1896 First modern > Olympic Games are held in Athens, Greece.

1900 William McKinley is reelected.

1901



The Expansion of Industry

The Big Idea

At the end of the 19th century, natural resources, creative ideas, and growing markets fueled an industrial boom.

Why It Matters Now

Technological developments of the late 19th century paved the way for the continued growth of American industry.

Key Terms and People

Edwin L. Drake

Bessemer process

Thomas Alva Edison

Lewis H. Latimer

Christopher Sholes

Alexander Graham Bell

One American's Story

One day, Pattillo Higgins noticed bubbles in the springs around Spindletop, a hill near Beaumont in southeastern Texas. This and other signs convinced him that oil was underground. If Higgins found oil, it could serve as a fuel source around which a vibrant industrial city would develop.

Higgins, who had been a mechanic and a lumber merchant, couldn't convince geologists or investors that oil was present, but he didn't give up. A magazine ad seeking investors got one response. It was from Captain Anthony F. Lucas, an experienced prospector who also believed that there was oil at Spindletop. When other investors were slow to send money, Higgins kept his faith, not only in Spindletop, but in Lucas.



Pattillo Higgins

"Captain Lucas, . . . these experts come and tell you this or that can't happen because it has never happened before. You believe there is oil here, . . . and I think you are right. I know there is oil here in greater quantities than man has ever found before."

—Pattillo Higgins, quoted in *Spindletop*

In 1900 the two men found investors, and they began to drill that autumn. After months of difficult, frustrating work, on the morning of January 10, 1901, oil gushed from their well. The Texas oil boom had begun.

Vocabulary
entrepreneur a person who organizes, operates, and assumes the risk for a business venture

Natural Resources Fuel Industrialization

During the early 1800s an industrial revolution brought new technology and industries to the United States. Factories and machines began to do much of the work that had previously been done by hand. By the end of the Civil War, however, the nation was still largely agricultural. In the late 1800s another period of industrial growth and technological development occurred in the United States. By the 1920s it had become the leading industrial power in the world. This immense industrial boom is sometimes called the **Second Industrial Revolution**. It was due to several factors, including new uses of natural resources, government support for business and innovation, and a growing urban population that provided both cheap labor and markets for new products.

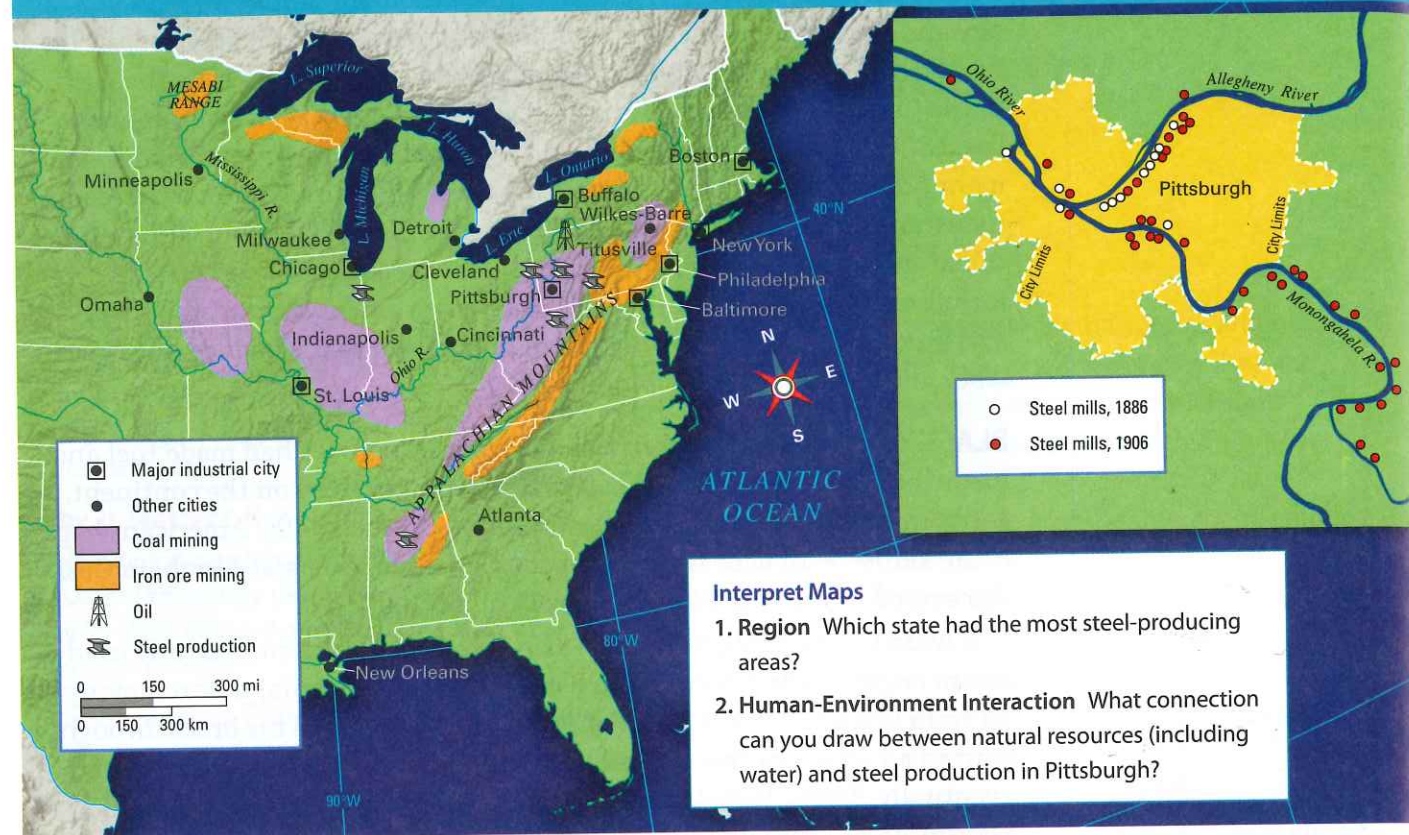
BLACK GOLD Though eastern Native American tribes had made fuel and medicine from crude oil long before Europeans arrived on the continent, early American settlers had little use for oil. In the 1840s, Americans began using kerosene to light lamps after the Canadian geologist Abraham Gesner discovered how to distill the fuel from oil or coal.

It wasn't until 1859, however, when **Edwin L. Drake** successfully used a steam engine to drill for oil near Titusville, Pennsylvania, that removing oil from beneath the earth's surface became practical. This breakthrough started an oil boom that spread to Kentucky, Ohio, Illinois, Indiana, and, eventually, Texas. Petroleum refining industries arose in Cleveland and Pittsburgh as entrepreneurs rushed to transform the oil into kerosene. Gasoline, a byproduct of the refining process, originally was thrown away. But after the automobile became popular, gasoline became the most important form of oil.

BESSEMER STEEL PROCESS Oil was not the only natural resource that was plentiful in the United States. There were also abundant deposits of coal and iron. In 1887 prospectors discovered iron ore deposits more than 100 miles long and up to 3 miles wide in the Mesabi Range of Minnesota. At the same time, coal production skyrocketed—from 33 million tons in 1870 to more than 250 million tons in 1900.

Iron is a dense metal, but it is soft and tends to break and rust. It also usually contains other elements, such as carbon. Removing the carbon from iron produces a lighter, more flexible, and rust-resistant metal—steel. The raw materials needed to make steel were readily available. All that was needed was a cheap and efficient manufacturing process. The **Bessemer process** was developed independently by both the British manufacturer Henry Bessemer and American ironmaker William Kelly around 1850. It soon became widely used. This technique involved injecting air into molten iron to remove the carbon and other impurities. By 1880 American manufacturers were using the new method to produce more than 90 percent of the nation's steel. In that age of rapid change and innovation, even the successful Bessemer process was improved by the 1860s. It was eventually replaced by the open-hearth process. This process enabled manufacturers to produce quality steel from scrap metal as well as from raw materials.

Natural Resources and the Birth of a Steel Town, 1886–1906



NEW USES FOR STEEL The railroads, with thousands of miles of track, became the biggest customers for steel, but inventors soon found additional uses for it. Cyrus McCormick and John Deere used steel to create mechanical farming equipment such as reapers and plows. This equipment helped transform the plains region into the food producer of the nation. These improvements in technology made farming more efficient, but it also meant that fewer laborers were needed to work the land. As a result, many people had to move from rural areas to cities to find work.

Steel changed the face of the nation as well. It made innovative construction possible. One of the most remarkable structures was the Brooklyn Bridge. Completed in 1883, it spanned 1,595 feet of the East River in New York City. Its steel cables were supported by towers higher than any man-made and weight-bearing structure except the pyramids of Egypt. Like those ancient marvels, the completed bridge was called a wonder of the world.

Around this time, William Le Baron Jenney designed the first skyscraper with a steel frame—the Home Insurance Building in Chicago. This set the stage for a new era of expansion upward as well as outward. Before Jenney had his pioneering idea, the weight of large buildings was supported entirely by their walls or by iron frames, which limited the buildings' height. With a steel frame to support the weight, however, architects could build as high as they wanted. As structures soared into the air, not even the sky seemed to limit what Americans could achieve.

Reading Check

Summarize
What natural resources were most important for industrialization?

Inventions Promote Change

By capitalizing on natural resources, their own ingenuity, and government support for innovation in the form of patents, inventors changed more than the landscape. Their inventions affected the very way people lived and worked.

THE POWER OF ELECTRICITY In 1876 **Thomas Alva Edison** became a pioneer on the new industrial frontier when he established the world's first research laboratory in Menlo Park, New Jersey. There Edison and his associates worked to perfect the incandescent light bulb, applying for a patent for his version of the bulb in 1879. **Lewis H. Latimer**, an African American inventor, played a key role in improving the light bulb when he invented a carbon filament in 1881. This filament lasted longer than filaments made from other materials that had been used in the past. Latimer later went to work for Edison's company. Eventually, Edison and his team improved the light bulb even further by using tungsten filaments, which are still used today.

Thomas Edison also invented an entire system for producing and distributing electrical power. Another inventor, George Westinghouse, along with Edison, added innovations that made electricity safer and less expensive. The harnessing of electricity completely changed the nature of business in America. By 1890 electric power ran numerous machines, from fans to printing presses. This inexpensive, convenient source of energy

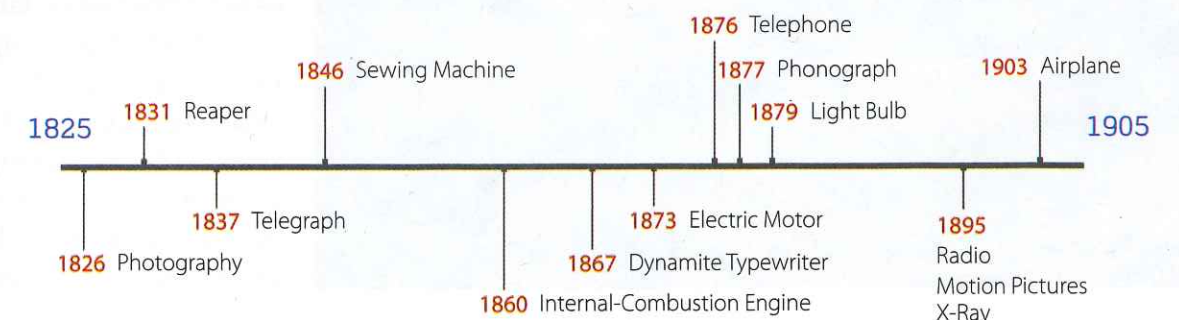
soon became available in homes and spurred the invention of time-saving appliances. Electric streetcars made urban travel cheap and efficient and also promoted the outward spread of cities.

More important, electricity allowed business owners greater freedom in deciding where to locate their plants. Many factories in the past, including textile mills, were powered by water. That meant they had to be located near rivers or streams. The availability of electric power meant that

Edison hoped to create a long-lasting light bulb with a soft glow.



The Technological Explosion, 1825–1905



manufacturers could consider other factors such as population and the availability of natural resources when determining where to locate their plants. Many automobile manufacturers, for example, chose to locate their plants in Detroit, Michigan. There rail and water shipping routes were easily accessible and coal and iron mines were nearby. Greater flexibility enabled industries to grow as never before.

INVENTIONS CHANGE LIFESTYLES Edison's light bulb was only one of several revolutionary inventions of the late 1800s. **Christopher Sholes** invented the typewriter in 1867 and changed the world of work. Next to the light bulb, however, perhaps the most dramatic invention was the telephone. **Alexander Graham Bell** and Thomas Watson unveiled this invention in 1876. It allowed people to talk to each other over long distances and opened the way for a worldwide communications network. Companies quickly found the telephone to be an essential business tool.

The typewriter and the telephone particularly affected office work and created new jobs for women. Businesses began to hire women as typists to manage company correspondence. Women made up less than 5 percent of all office workers in 1870. By 1910 they accounted for nearly 40 percent of the clerical work force.

The typewriter shown here dates from around 1890.



New inventions also had a tremendous impact on factory work, as well as on jobs that traditionally had been done at home. For example, women had previously sewn clothing by hand for their families. With industrialization, clothing could be mass-produced in factories. This created a need for garment workers, many of whom were women.

Industrialization freed some factory workers from backbreaking labor and helped make their jobs easier. Shoemaker Jan Ernst Matzeliger designed a machine that could attach the upper part of shoes to the soles, helping to automate the shoe manufacturing process. Margaret E. Knight invented

several machines that made the shoemaking process more efficient. Matzeliger's and Knight's contributions to the footwear industry helped lower the cost of production. Automation made it possible to make larger quantities of shoes in a shorter amount of time.

By 1890 the average workweek in the United States had been reduced by about ten hours. However, many laborers felt that the mechanization of so many tasks reduced human workers' worth. As consumers, though, workers regained some of their lost power in the marketplace.

The inventions and processes of the late 1800s and early 1900s increased the standard of living in the United States and created notable wealth for business owners. The increased living standards attracted new immigrants seeking opportunities for a better life.

The rise of immigration brought a new set of challenges and concerns to the democratic system of the United States. It also brought new workers and potential consumers to the economy. The country's expanding urban population provided a vast potential market for the new inventions and products of the late 19th century.

Reading Check
Analyze Effects How did electricity change American life?

Lesson 1 Assessment

1. **Organize Information** Use a chart to list resources, ideas, and markets that affected the industrial boom of the 19th century. In the second column, note how each item contributed to industrialization.

Resources, Ideas, Markets	Impact

2. **Key Terms and People** For each key term or person in the lesson, write a sentence explaining its significance.

3. **Analyze Effects** Which invention or development described in this lesson had the greatest impact on society? Justify your choice.

Think About:

- the applications of inventions
- the impact of inventions on people's daily lives
- the effect of inventions on the workplace

4. **Make Inferences** Do you think consumers gained power as industry expanded in the late 19th century? Why or why not?

5. **Hypothesize** If the United States had been poor in natural resources, how would industrialization have been affected?